SATA TSG, RSG Test Report

Overall result: Pass

DUT:	A9126_new
Comment:	PCI 33M
Time of session start:	03/25/2014 15:08:30
Operator:	
Temperature	26° C
Standard in use:	SATA TSG, RSG

Run1:

Time of run:	2014/03/25 15:08:31
Configuration in use:	Gen2 (3.0Gb/s) Device (Copy)
Limits in use:	Gen2i
Oscilloscope Name:	LCRY0433N66554 Model: SDA813ZI-A
Oscilloscope Serial #:	LCRY0433N66554
Computer:	LCRY0433N66554
Oscilloscope firmware version:	7.1.1.2 (Build 184204)
QualiPHY core version:	7.0.4.4 (Build 184490)
QualiPHY script version:	7.0.4.4
Stylesheet version:	1.2.0.6
MOI_Revision:	1.4.3
Waveform path on oscilloscope	D:\Waveforms\SATA\A9126_new

Summary Table

[Hide Table]

Pass	Run #	Test	Measurement	Current Value	Test Criteria
ℹ	1	PHY-01	Unit Interval Min	333.1989 ps	Informational Only
ℹ	1	PHY-01	Unit Interval Max	333.4007 ps	Informational Only
✔	1	PHY-01	Unit Interval Mean	333.2979 ps	333.2167 ps <= x <= 335.1167 ps
✔	1	PHY-02	ftol	106.42 ppm	-350.00 ppm <= x <= 350.00 ppm
ℹ	1	TSG-01	UH HFTP	247.0 mV	Informational Only
ℹ	1	TSG-01	LH HFTP	-260.2 mV	Informational Only
✔	1	TSG-01	DH HFTP	507.2 mV	x >= 400.0 mV
ℹ	1	TSG-01	UM MFTP	236.8 mV	Informational Only
ℹ	1	TSG-01	LM MFTP	-242.1 mV	Informational Only
✔	1	TSG-01	DM MFTP	479.0 mV	x >= 400.0 mV
✔	1	TSG-01	DHM	479.0 mV	x >= 400.0 mV
ℹ	1	TSG-01	A LBP	224.8 mV	Informational Only
ℹ	1	TSG-01	B LBP	-233.9 mV	Informational Only
✔	1	TSG-01	VTestLBP	458.7 mV	x >= 400.0 mV
✔	1	TSG-01	VTest	458.7 mV	x >= 400.0 mV
ℹ	1	TSG-01	pu MFTP	0.0 m	Informational Only
ℹ	1	TSG-01	pl MFTP	0.0 m	Informational Only
ℹ	1	TSG-01	pu LFTP	0.0 m	Informational Only
ℹ	1	TSG-01	pl LFTP	0.0 m	Informational Only
ℹ	1	TSG-02	TX Rise HFTP	85.856 ps	Informational Only
ℹ	1	TSG-02	TX Fall HFTP	85.107 ps	Informational Only
✔	1	TSG-02	TX Rise LFTP	88.545 ps	x <= 136.000 ps
✔	1	TSG-02	TX Fall LFTP	88.237 ps	x <= 136.000 ps
✔	1	TSG-03	TskewTX HFTP	797 fs	x <= 20.0 ps
✔	1	TSG-03	TskewTX MFTP	142 fs	x <= 20.0 ps
✔	1	TSG-04	Vcm,acTX	10.8 mV	x <= 50.0 mV
✔	1	TSG-11	Tj fBAUD/500 HFTP	125 mUI	x <= 370 mUI
✔	1	TSG-11	Tj fBAUD/500 LBP	236 mUI	x <= 370 mUI
ℹ	1	TSG-11	Tj fBAUD/500 SSOP	206 mUI	Informational Only
ℹ	1	TSG-11	Tj (1e-6) HFTP	90 mUI	Informational Only
ℹ	1	TSG-11	Tj (1e-6) LBP	181 mUI	Informational Only

	1	TSG-11	Tj (1e-6) SSOP	159 mUI	Informational Only
	1	TSG-12	Dj fBAUD/500 HFTP	14 mUI	x <= 190 mUI
	1	TSG-12	Dj fBAUD/500 LBP	67 mUI	x <= 190 mUI
	1	TSG-12	Dj fBAUD/500 SSOP	61 mUI	Informational Only

Details

Phy General Requirements

Test PHY-01 - Unit Interval

[Up]

	Measurement:	***Unit Interval Min***
	Limit Name:	InfoOnlyPhy01
	Current Value:	333.1989 ps
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:09:48
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

	Measurement:	***Unit Interval Max***
	Limit Name:	InfoOnlyPhy01
	Current Value:	333.4007 ps
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:09:48
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

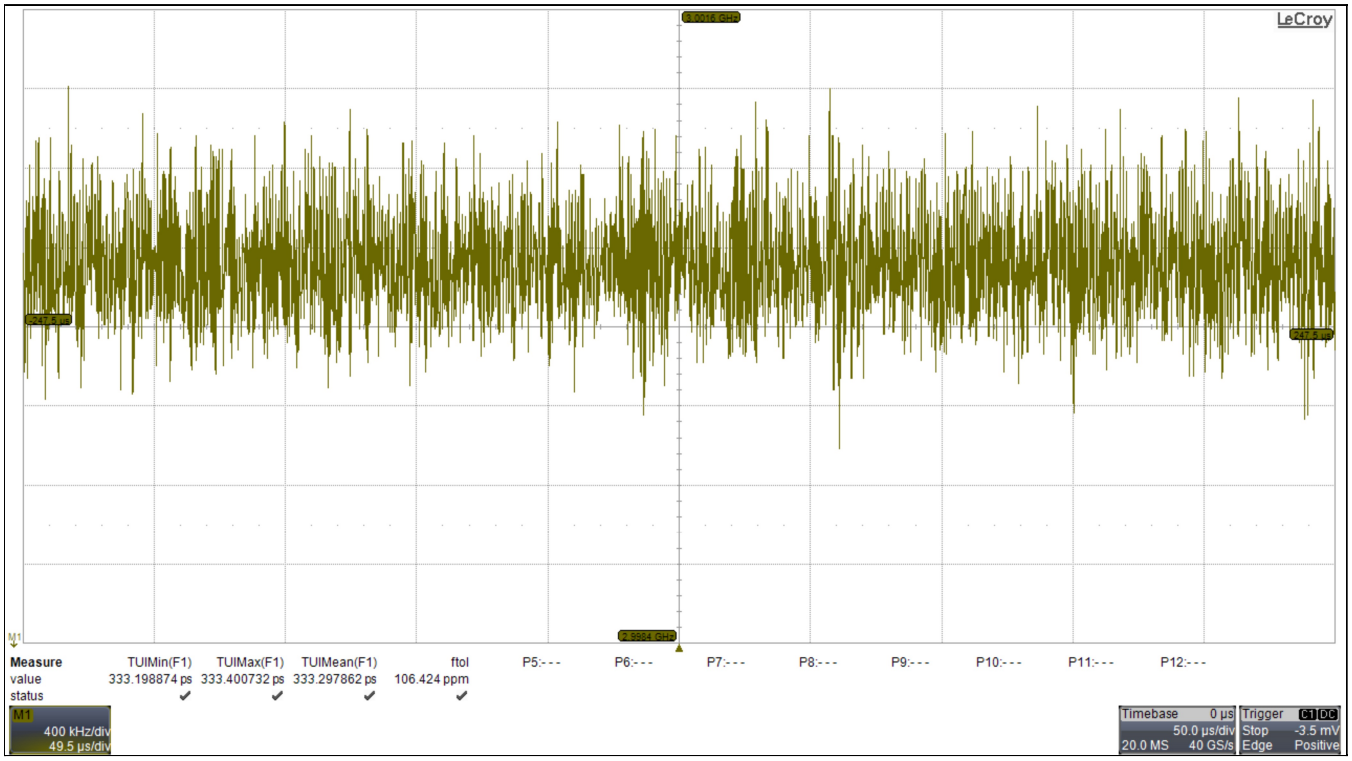
 **Pass**	Measurement:	***Unit Interval Mean***
	Limit Name:	UnitInterval
	Current Value:	333.2979 ps
	Test Criteria:	333.2167 ps <= x <= 335.1167 ps
	Timestamp:	03/25/2014 15:09:48
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

Test PHY-02 - Frequency Long Term Stability

[Up]

 **Pass**	Measurement:	***ftol***
	Limit Name:	FrequencyLongTermStability
	Current Value:	106.42 ppm
	Test Criteria:	-350.00 ppm <= x <= 350.00 ppm
	Timestamp:	03/25/2014 15:09:49

Run Number	1
TXnWaveform	N/A
TXpWaveform	N/A
pattern	HFTP



Phy General Requirements

Timestamp: 03/25/2014 15:09:49

"Phy Transmitted Signal Requirements"

Test TSG-01 - Differential Output Voltage HFTP

[Up]

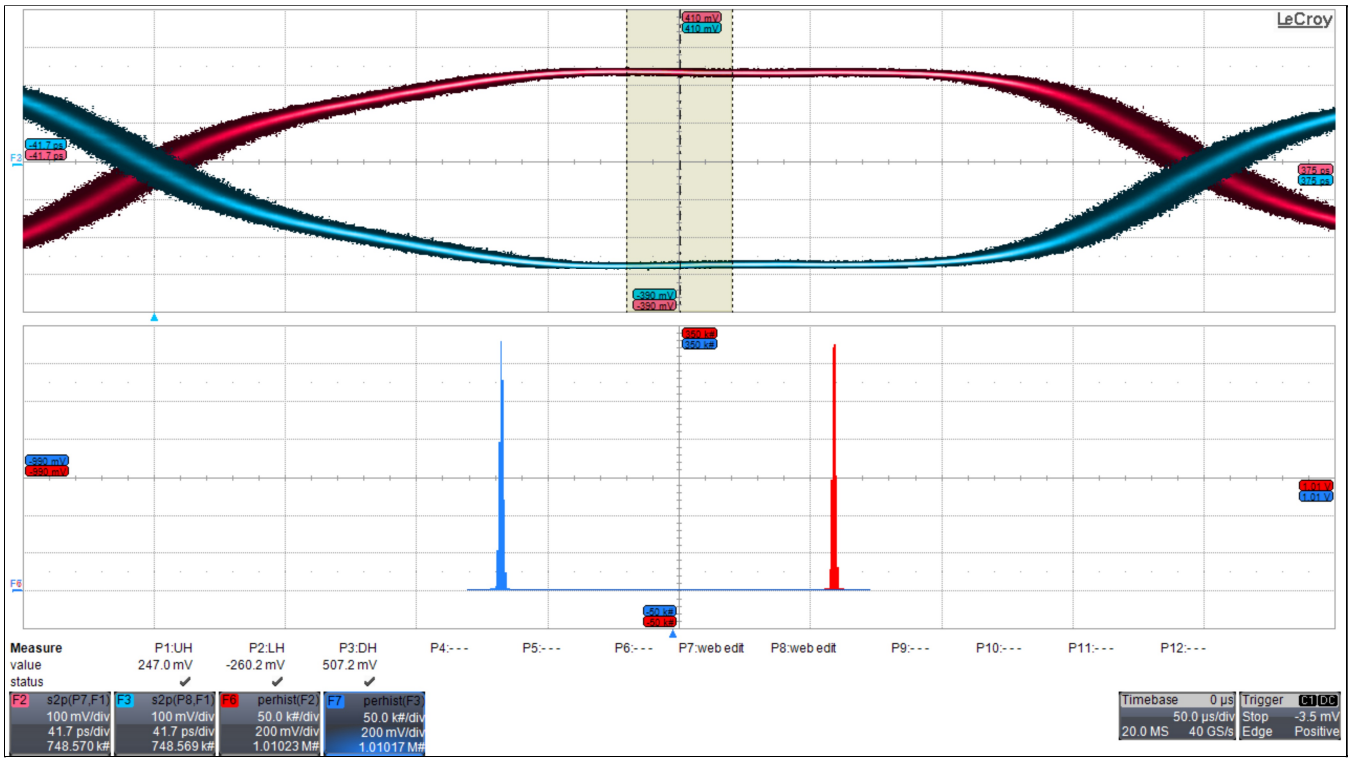
	Measurement:	***UH HFTP***
	Limit Name:	InfoOnlyV
	Current Value:	247.0 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:10:03
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

	Measurement:	***LH HFTP***
	Limit Name:	InfoOnlyV
	Current Value:	-260.2 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:10:03
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

 **Pass**	Measurement:	***DH HFTP***
	Limit Name:	VdiffMin
	Current Value:	507.2 mV
	Test Criteria:	x >= 400.0 mV
	Timestamp:	03/25/2014 15:10:03
	Run Number	1



TSG-01: VDiffMin HFTP

Timestamp: 03/25/2014 15:10:03

Test TSG-01 - Differential Output Voltage MFTP

[Up]

	Measurement:	***UM MFTP***
	Limit Name:	InfoOnlyV
	Current Value:	236.8 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:11:05
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP

[Up]

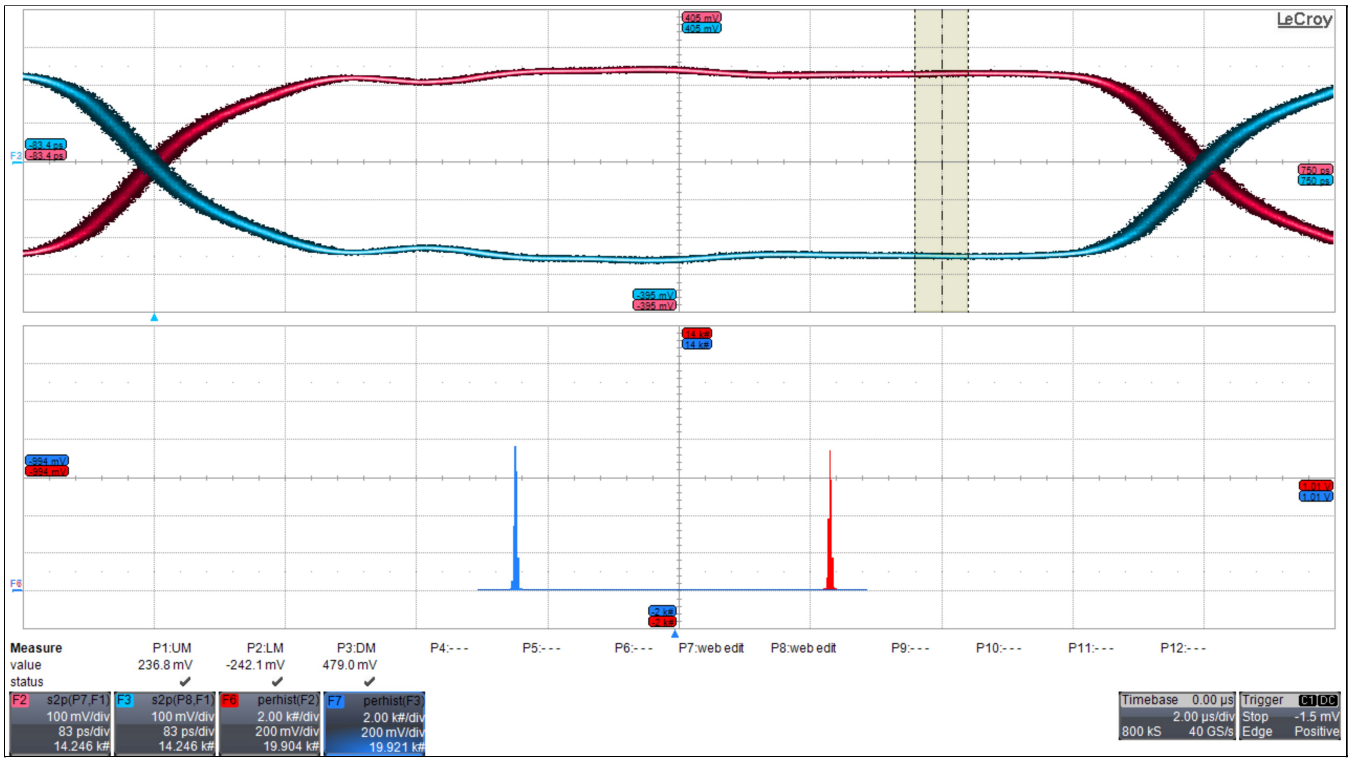
	Measurement:	***LM MFTP***
	Limit Name:	InfoOnlyV
	Current Value:	-242.1 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:11:05
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP

[Up]

 **Pass**	Measurement:	***DM MFTP***
	Limit Name:	VdiffMin
	Current Value:	479.0 mV
	Test Criteria:	x >= 400.0 mV
	Timestamp:	03/25/2014 15:11:05
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP

[Up]

 **Pass**	Measurement:	***DHM***
	Limit Name:	VdiffMin
	Current Value:	479.0 mV
	Test Criteria:	x >= 400.0 mV
	Timestamp:	03/25/2014 15:11:05
	Run Number	1



TSG-01: VDiffMin MFTP

Timestamp: 03/25/2014 15:11:05

Test TSG-01 - Differential Output Voltage LBP

[Up]

	Measurement:	***A LBP***
	Limit Name:	InfoOnlyV
	Current Value:	224.8 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:13:15
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

[Up]

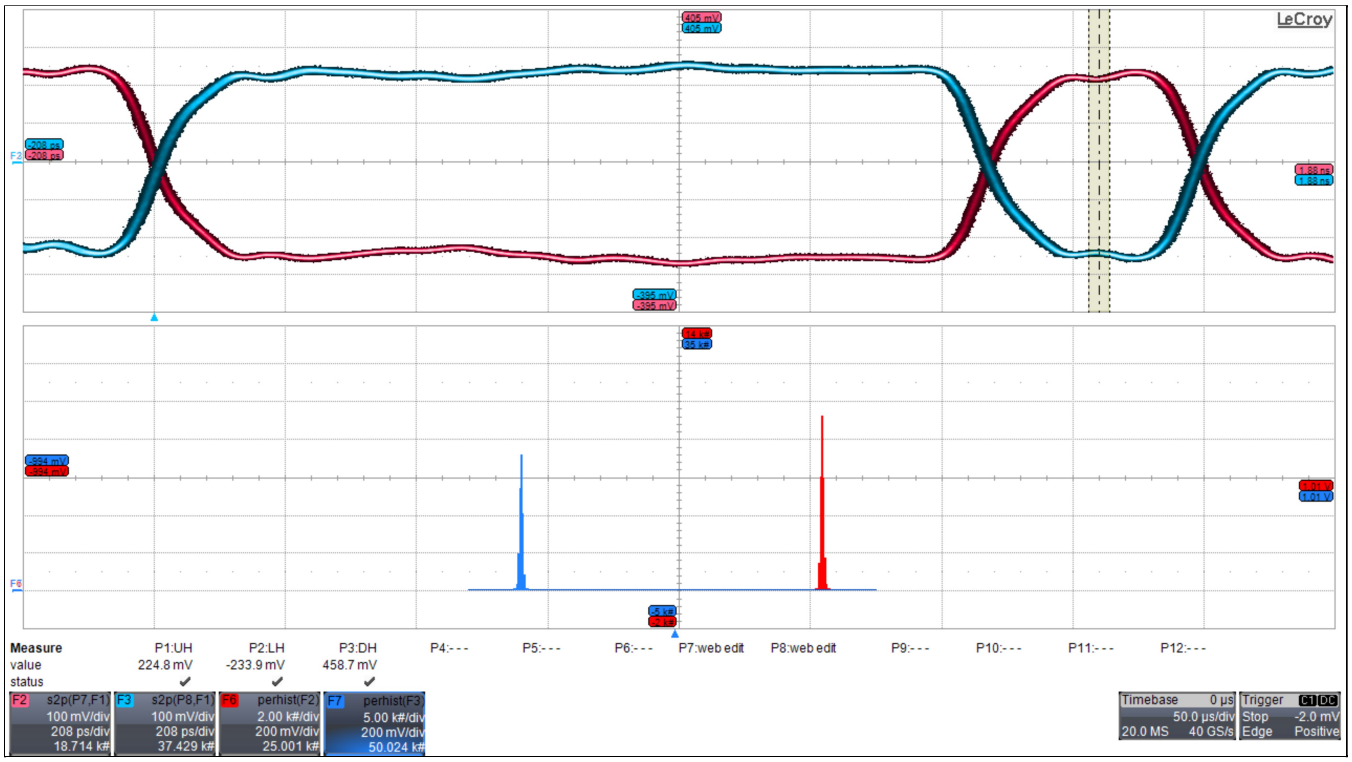
	Measurement:	***B LBP***
	Limit Name:	InfoOnlyV
	Current Value:	-233.9 mV
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:13:15
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

[Up]

 Pass	Measurement:	***VTestLBP***
	Limit Name:	VdiffMin
	Current Value:	458.7 mV
	Test Criteria:	x >= 400.0 mV
	Timestamp:	03/25/2014 15:13:15
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

[Up]

 Pass	Measurement:	***VTest***
	Limit Name:	VdiffMin
	Current Value:	458.7 mV
	Test Criteria:	x >= 400.0 mV
	Timestamp:	03/25/2014 15:13:15
	Run Number	1



TSG-01: VDiffMin LBP

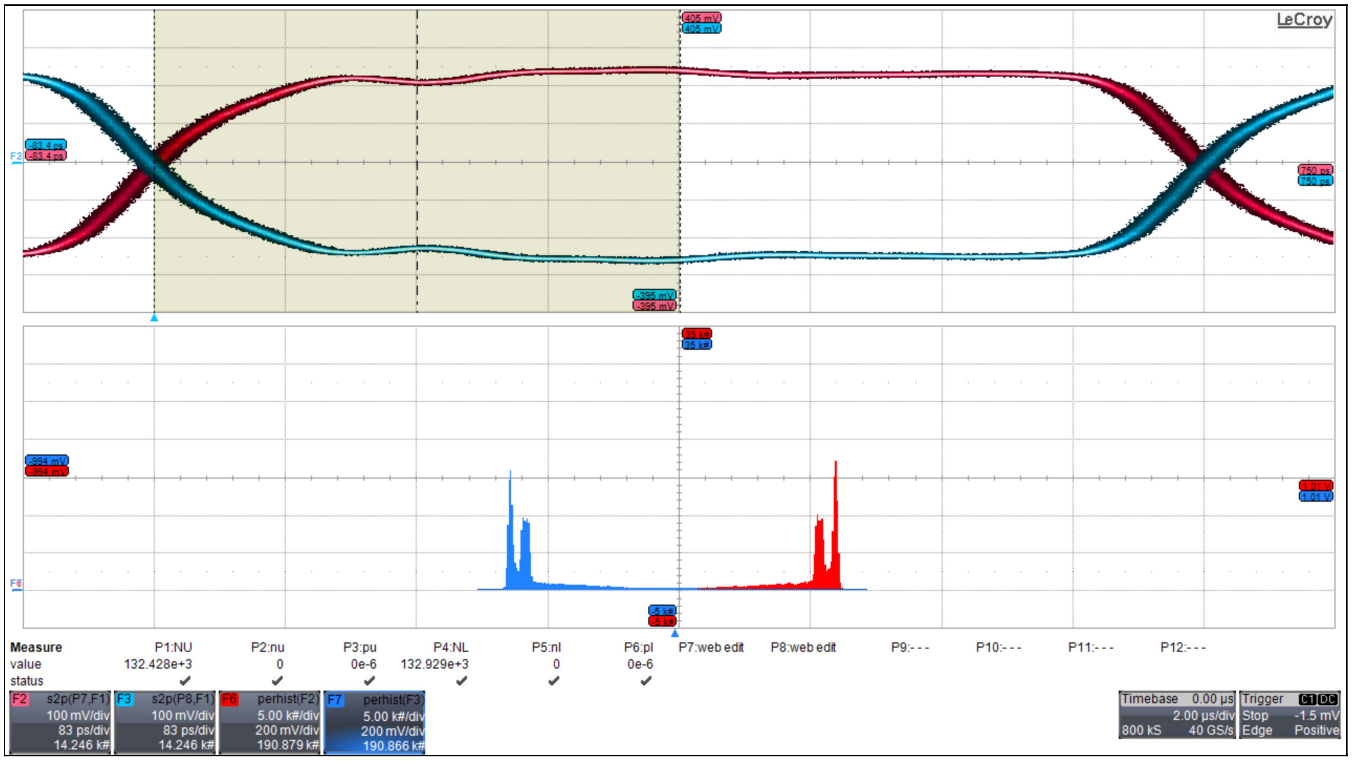
Timestamp: 03/25/2014 15:13:15

[Up]

	Measurement:	***pu MFTP***
	Limit Name:	pu/pl
	Current Value:	0.0 m
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:11:10
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP

[Up]

	Measurement:	***pl MFTP***
	Limit Name:	pu/pl
	Current Value:	0.0 m
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:11:10
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP



TSG-01: VdiffMax MFTP

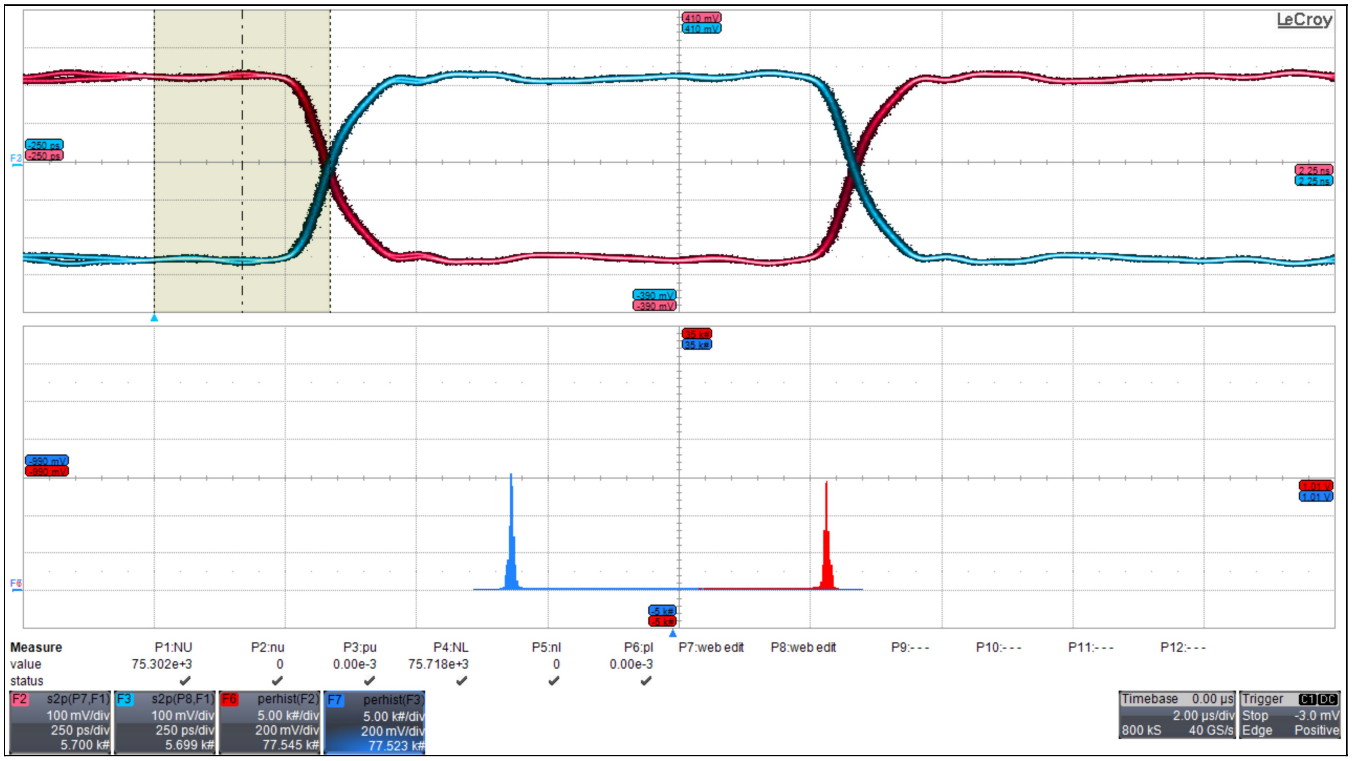
Timestamp: 03/25/2014 15:11:11

[Up]

	Measurement:	***pu LFTP***
	Limit Name:	pu/pl
	Current Value:	0.0 m
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:12:21
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LFTP

[Up]

	Measurement:	***pl LFTP***
	Limit Name:	pu/pl
	Current Value:	0.0 m
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:12:21
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LFTP



TSG-01: VdiffMax LFTP

Timestamp: 03/25/2014 15:12:22

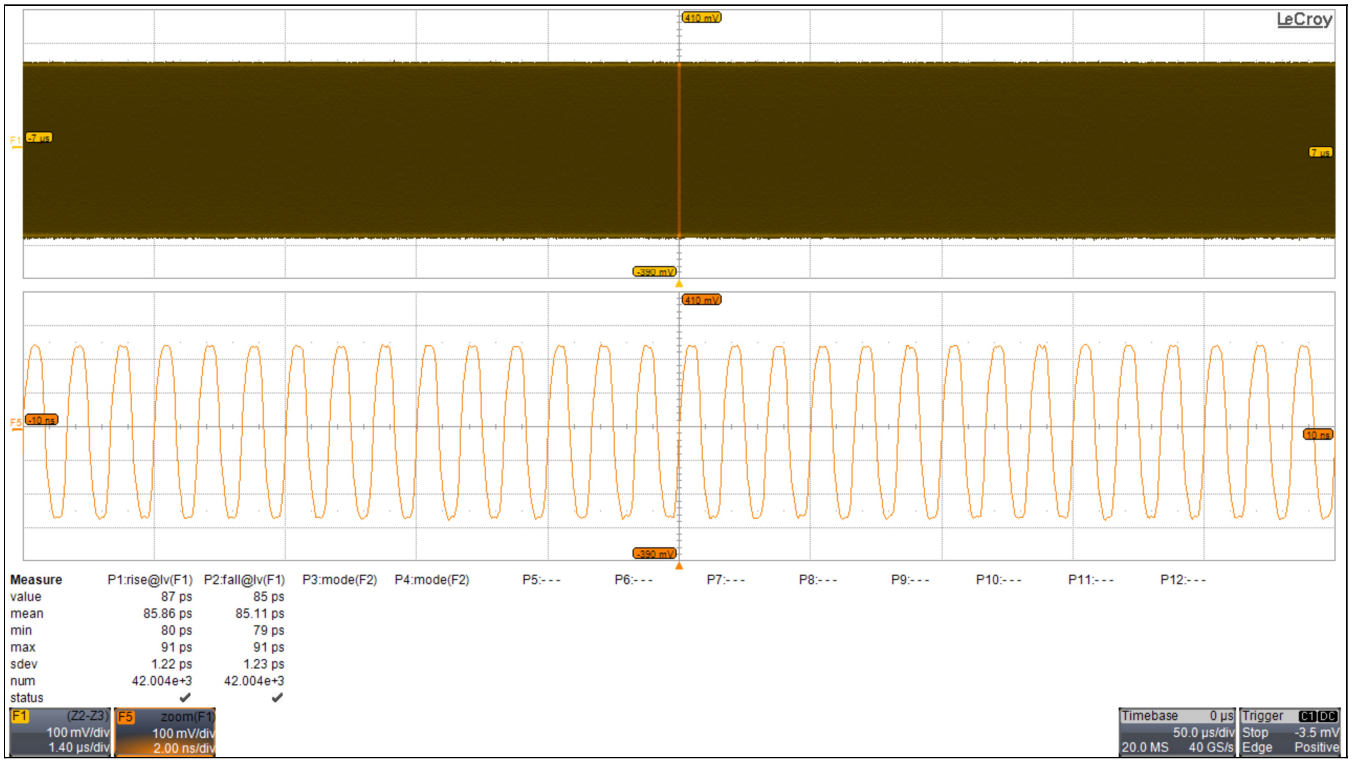
Test TSG-02 - Rise/Fall Time HFTP

[Up]

	Measurement:	***TX Rise HFTP***
	Limit Name:	InfoOnlySec
	Current Value:	85.856 ps
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:10:14
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

	Measurement:	***TX Fall HFTP***
	Limit Name:	InfoOnlySec
	Current Value:	85.107 ps
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:10:14
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP



TSG-02: Rise/Fall Time

Timestamp: 03/25/2014 15:10:15

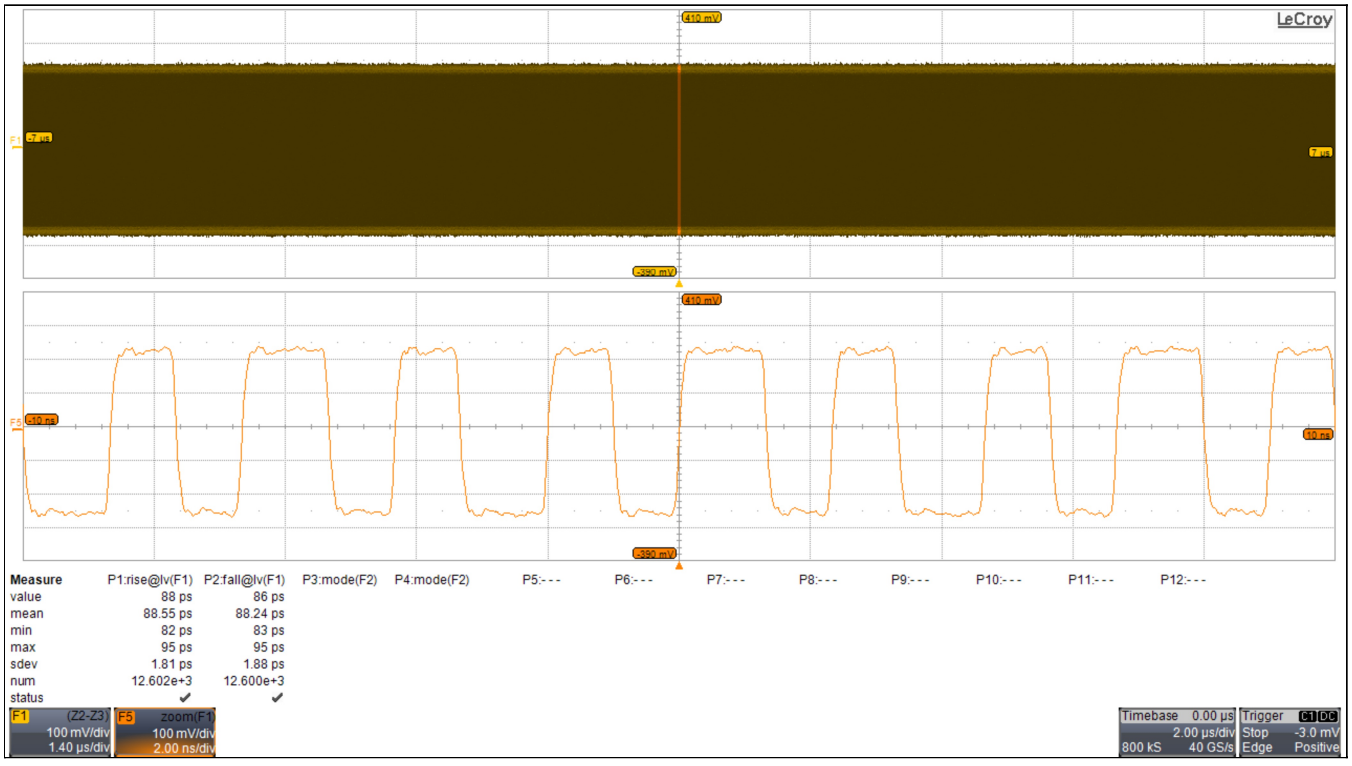
Test TSG-02 - Rise/Fall Time LFTP

[Up]

✔ Pass	Measurement:	***TX Rise LFTP***
	Limit Name:	Rise/FallTime
	Current Value:	88.545 ps
	Test Criteria:	x <= 136.000 ps
	Timestamp:	03/25/2014 15:12:26
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LFTP

[Up]

✔ Pass	Measurement:	***TX Fall LFTP***
	Limit Name:	Rise/FallTime
	Current Value:	88.237 ps
	Test Criteria:	x <= 136.000 ps
	Timestamp:	03/25/2014 15:12:26
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LFTP



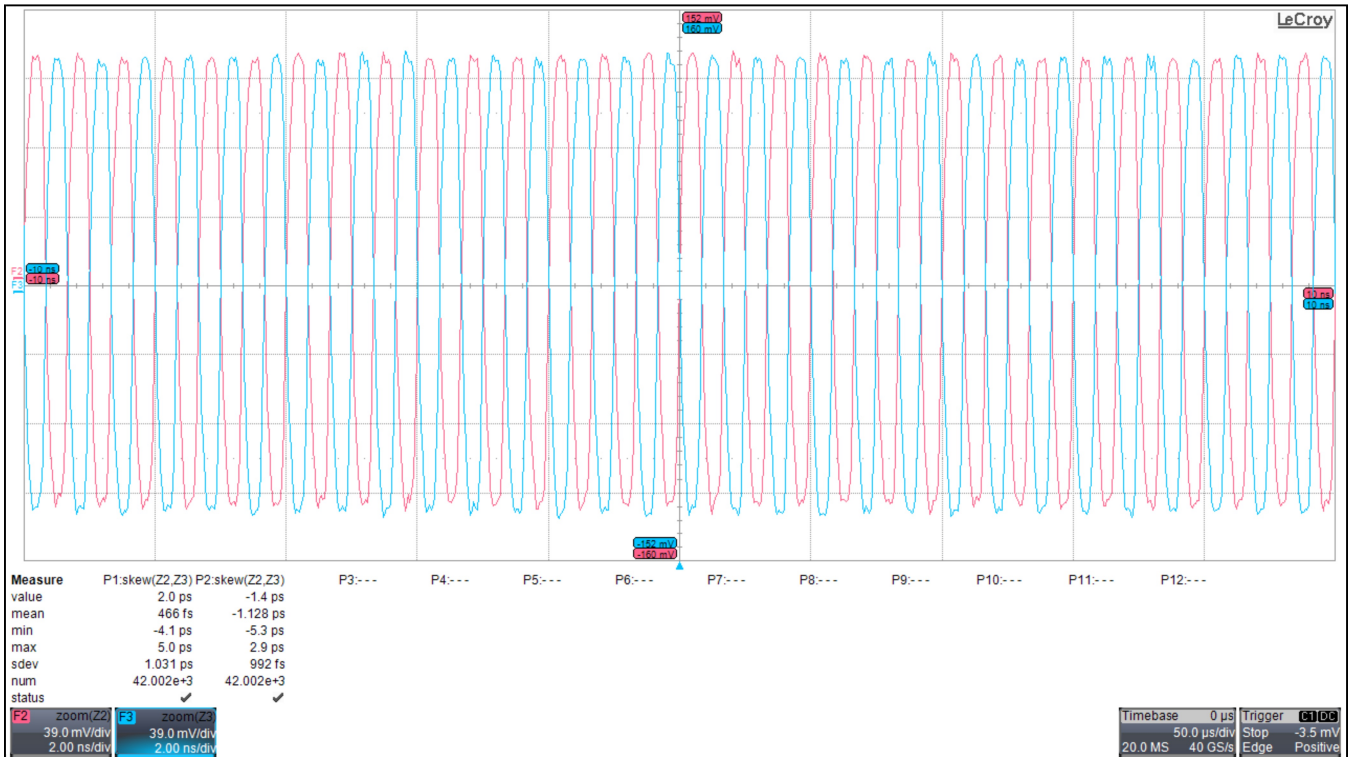
TSG-02: Rise/Fall Time

Timestamp: 03/25/2014 15:12:27

Test TSG-03 - Differential Skew HFTP

[Up]

✔ Pass	Measurement:	***TskewTX HFTP***
	Limit Name:	DifferentialSkew
	Current Value:	797 fs
	Test Criteria:	x <= 20.0 ps
	Timestamp:	03/25/2014 15:10:19
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP



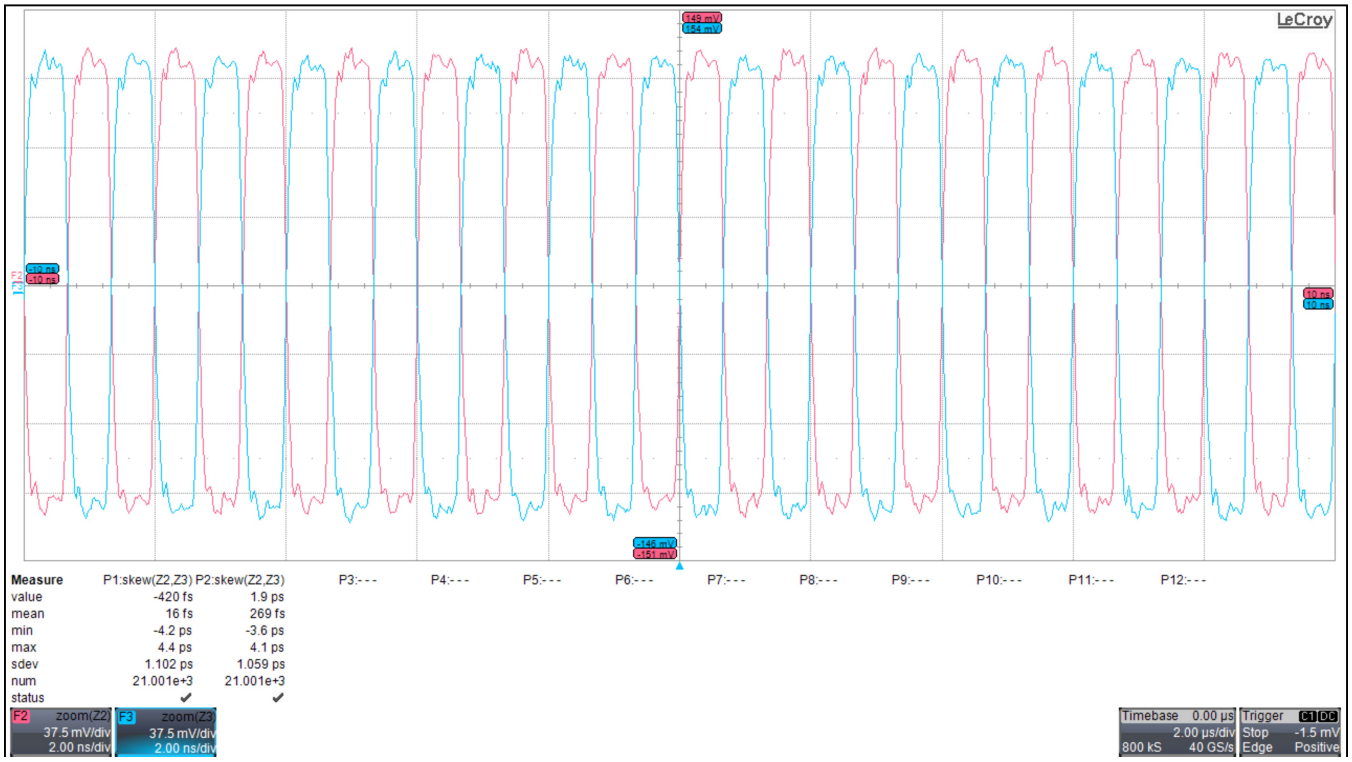
TSG-03: Differential Skew

Timestamp: 03/25/2014 15:10:20

Test TSG-03 - Differential Skew MFTP

[Up]

✔ Pass	Measurement:	***TskewTX MFTP***
	Limit Name:	DifferentialSkew
	Current Value:	142 fs
	Test Criteria:	x <= 20.0 ps
	Timestamp:	03/25/2014 15:11:16
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP



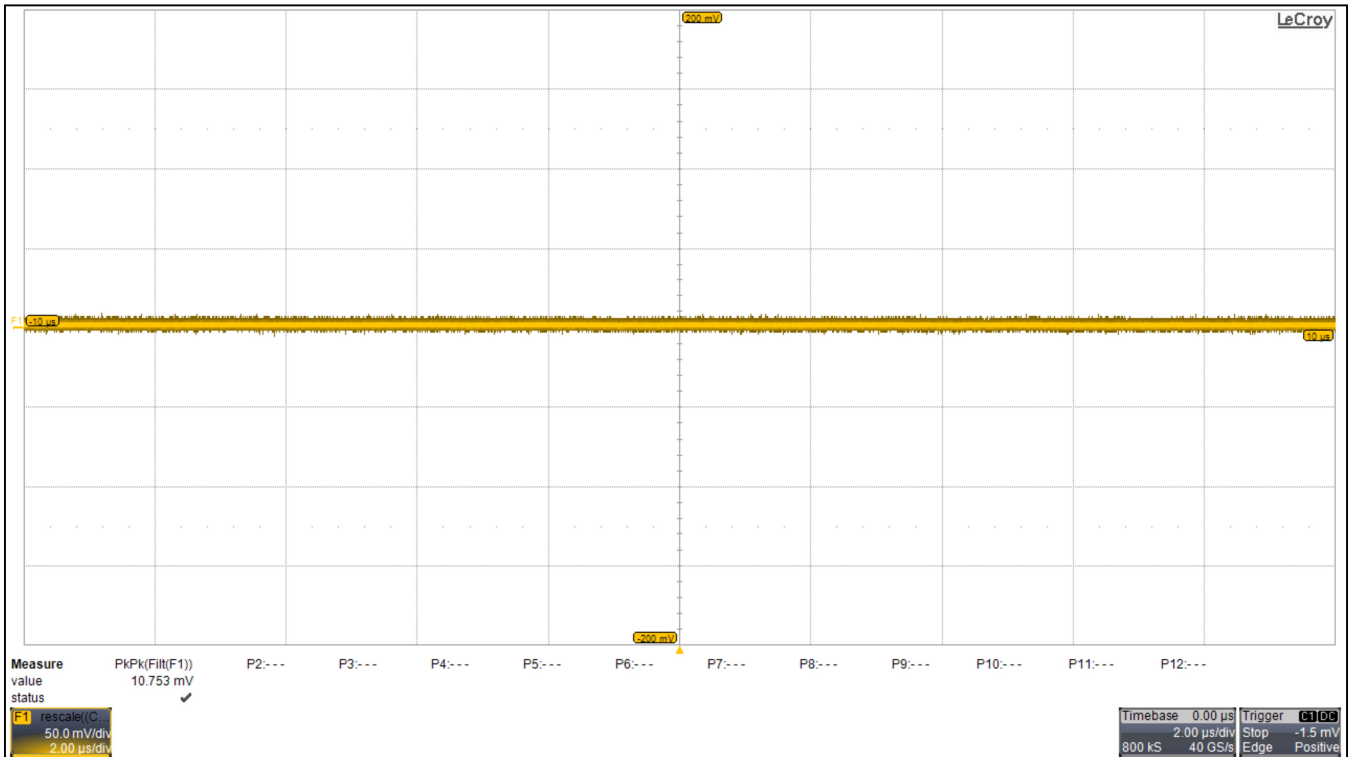
TSG-03: Differential Skew

Timestamp: 03/25/2014 15:11:16

Test TSG-04 - AC Common Mode Voltage

[Up]

✔ Pass	Measurement:	***Vcm,acTX***
	Limit Name:	ACCMVoltage
	Current Value:	10.8 mV
	Test Criteria:	x <= 50.0 mV
	Timestamp:	03/25/2014 15:11:18
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	MFTP



TSG-04: AC Common Mode Voltage

Timestamp: 03/25/2014 15:11:18

Test TSG-11 - Gen2 Tj fBAUD/500, Clock to Data HFTP

[Up]

Pass	Measurement:	***Tj fBAUD/500 HFTP***
	Limit Name:	TJ500
	Current Value:	125 mUI
	Test Criteria:	x <= 370 mUI
	Timestamp:	03/25/2014 15:10:27
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

Test TSG-11 - Gen2 Tj fBAUD/500, Clock to Data LBP

[Up]

Pass	Measurement:	***Tj fBAUD/500 LBP***
	Limit Name:	TJ500
	Current Value:	236 mUI
	Test Criteria:	x <= 370 mUI
	Timestamp:	03/25/2014 15:13:28
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

Test TSG-11 - Gen2 Tj fBAUD/500, Clock to Data SSOP

[Up]

	Measurement:	***Tj fBAUD/500 SSOP***
	Limit Name:	InformationalOnlyUI
	Current Value:	206 mUI
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:14:17
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	SSOP

[Up]

	Measurement:	***Tj (1e-6) HFTP***
	Limit Name:	InformationalOnlyUI
	Current Value:	90 mUI
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:10:28
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

[Up]

	Measurement:	***Tj (1e-6) LBP***
	Limit Name:	InformationalOnlyUI
	Current Value:	181 mUI
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:13:29
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

[Up]

	Measurement:	***Tj (1e-6) SSOP***
	Limit Name:	InformationalOnlyUI
	Current Value:	159 mUI
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:14:18
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	SSOP

Test TSG-12 - Gen2 Dj fBAUD/500, Clock to Data HFTP

[Up]

	Measurement:	***Dj fBAUD/500 HFTP***
	Limit Name:	DJ500
	Current Value:	14 mUI
	Test Criteria:	x <= 190 mUI
	Timestamp:	03/25/2014 15:10:27
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	HFTP

Test TSG-12 - Gen2 Dj fBAUD/500, Clock to Data LBP

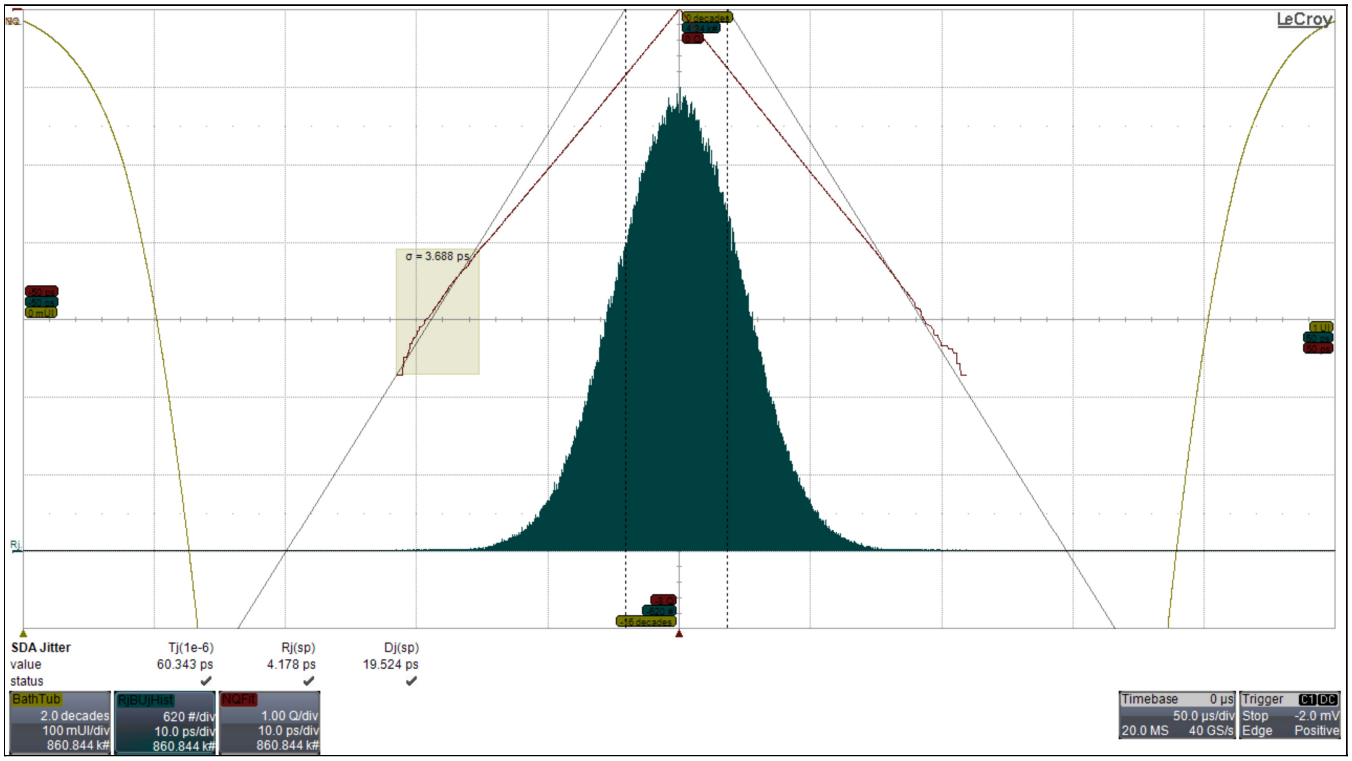
[Up]

	Measurement:	***Dj fBAUD/500 LBP***
	Limit Name:	DJ500
	Current Value:	67 mUI
	Test Criteria:	x <= 190 mUI
	Timestamp:	03/25/2014 15:13:28
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	LBP

Test TSG-12 - Gen2 Dj fBAUD/500, Clock to Data SSOP

[Up]

	Measurement:	***Dj fBAUD/500 SSOP***
	Limit Name:	InformationalOnlyUI
	Current Value:	61 mUI
	Test Criteria:	Informational Only
	Timestamp:	03/25/2014 15:14:18
	Run Number	1
	TXnWaveform	N/A
	TXpWaveform	N/A
	pattern	SSOP



TSG-11 and TSG-12: Gen2 Clock to Data Jitter, fBAUD/500, LBP

Timestamp: 03/25/2014 15:13:29

--- End of report ---